Benjamin Greenberg

Mechanical Engineering & Embedded Systems Engineering
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EDUCATION

Swarthmore College | Swarthmore, PA May 2029
Candidate for B.S. in Engineering and B.A. in Economics | GPA: 3.93/4.00 | Sigma Xi

ENGINEERING EXPERIENCE

Haptic Robotics Research Assistant | Swarthmore College October 2025 – Present
- Configuring a PHANToM Premium device for post-stroke rehabilitation research in a neuromuscular hand control lab
- Designing and 3D-printing a fingertip attachment for single-finger testing instead of whole-hand gripping
- Building 7 interactive 3D haptic apps in C++ using OpenHaptics SDK, increasing data capture from 0.2 Hz to 1 kHz
- Planning a study to measure whether post-stroke patients lose both feeling and movement control in the hand

Computational Network Dynamics Researcher | Lehigh University June 2025 – Present
- Built and ran an agent-based model of vaccination decisions and disease spread (1,000 agents, 30 replicates, tested up to 16,000), showing a small drop in social influence abruptly collapses high coverage, which then fails to recover.
- Authored a first-author manuscript on vaccination bistability, currently under review

Aerial Robotics Researcher | Lehigh University May 2026 – August 2026
- Engineered an autonomous blimp drone for rehabilitation applications in the Aerial Swarms Laboratory
- Designed and 3D-printed a custom drone frame integrating an ESP32 microcontroller and IMU, achieving a 29g total system weight to minimize the motor power needed for lift, iterating the frame design over 10+ builds
- Wrote a pure-pursuit & PD autonomous controller in C, returning the blimp toward target after all 23 hand pushes of up to 4.1 m, with 78% settling within 0.8 m and a 15 s median to remove 80% of the offset, under motion capture
- Second author on a human-drone intent recognition paper presented at IROS 2026 workshop, with a preprint on arXiv

PROJECTS

SEPTA Display May 2026 – Present
- Iterating a 16-panel HUB75 LED departure board (Raspberry Pi 5, 5V/100A) left as a rough senior design prototype
- Rebuilt the Python display layer with live train position from SEPTA TrainView GPS
- Redesigning the enclosure for weatherproof mounting and coordinating station installation with SEPTA

Cantilever Truss | ENGR 006 April 2026
- Designed, machined, and load-tested a 6.45 lb aluminum cantilever truss that carried a 1,000 lb point load
- Traced a 3x deflection error (0.12 in predicted, 0.39 in measured) to joint misalignment, not member sizing

Macropad May 2026
- Designed schematic and PCB in KiCad for a 9-key macropad, manually soldered it, and flashed custom KMK firmware

MakerWorld December 2024
- Published 10 3D-printable models in Fusion 360, drawing 5M+ views and 22K+ downloads to date

SKILLS & COURSEWORK

Tools & Lab: Soldering, continuity testing, FDM 3D printing, tensile testing, Raspberry Pi, ESP32 integration

Software & CAD: C, C++, Python, MATLAB, KiCad, SolidWorks, Autodesk Fusion, ROS

Coursework: Circuit Analysis, Differential Equations, Mechanics of Solids (in progress, Fall 2026); Linear Systems Analysis (expected, Spring 2027); Electricity and Magnetism, Engineering Static Mechanics, Multivariable Calculus (Completed)

Language Skills: French (fluent)

LEADERSHIP EXPERIENCE

Swarthmore Robotics Club, *President* February 2026 – Present
- Leading an 8 person team developing an F1TENTH autonomous racing car, integrating LiDAR SLAM and ROS navigation

Swarthmore College Admissions, *Campus Tour Guide* November 2025 – Present
- Guiding 60 minute campus tours for several hundred prospective students and families each year

Video & Graphic Design Business, *Founder* September 2020 – Present
- Founded and grew an online Minecraft art business to 55,000+ YouTube subscribers and 86,000+ TikTok followers

Swarthmore Clubs: Club Squash (Captain), Ceramics Club (Founder), Chabad (Board Member), 180 Degrees Consulting